

Aya Abdel Nasser Junior Data Engineer

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PROFESSIONAL SUMMARY

Junior Data Engineer with hands-on experience designing and implementing end-to-end data pipelines using Apache Spark, Kafka, Airflow, dbt, Docker, SQL Server, and BigQuery. Skilled in ETL/ELT development, data warehousing, batch and streaming data processing, and workflow orchestration. Built production-style projects involving real-time data ingestion, dimensional modeling, data quality validation, observability, and analytics dashboards. Focused on building scalable and reliable data platforms.

EXPERIENCE

01/2026 – 07/2026 | Cairo, Egypt

Data Engineer Trainee | Digital Pioneers Initiative (MCIT)

- Completed an intensive Data Engineering training program.
- Built batch and streaming data pipelines using Python and SQL.
- Applied ETL, dimensional modeling, and data quality techniques.
- Developed workflows using Spark, Airflow, Docker, and SQL Server.
- Collaborated with teams using Git and GitHub.

EDUCATION

Data Engineering Track — Digital Pioneers Initiative (MCIT), 2026

Bachelor of Arts and Education, Ain Shams University

CERTIFICATIONS

- Microsoft Certified: Fabric Data Engineer Associate (DP-700)
- Microsoft Certified: Azure Data Fundamentals (DP-900)

PROJECTS

PharmStock AI Platform V2 — End-to-End Data Engineering & MLOps Platform

- Architected and implemented an end-to-end data platform for pharmacy operations, integrating **CDC streaming, distributed processing, cloud analytics, MLOps, and BI workflows**.
- Built a modern data pipeline using **PostgreSQL → Debezium CDC → Apache Kafka → Apache Spark → Google BigQuery → dbt → Apache Airflow**.
- Developed analytics-ready data models and governed datasets supporting ML workflows, operational analytics, and Power BI reporting.
- Implemented MLOps workflow with **MLflow model tracking, registry, versioning, quality gates, and model serving lifecycle**.
- Applied data quality validation, orchestration, monitoring, RBAC, and human-in-the-loop decision governance.

Tech Stack: Python, SQL, PostgreSQL, Debezium, Apache Kafka, Apache Spark, BigQuery, dbt, Airflow, MLflow, FastAPI, Docker, Power BI.

GitHub: <https://github.com/Aya-Abdel-Nasser/pharmstock-ai-platform> 📄

End-to-End Sales Data Warehouse, ETL Pipeline

- Designed and implemented an end-to-end ETL pipeline using SQL Server Integration Services (SSIS) to populate a Sales Data Warehouse from OLTP source systems.
- Built a Star Schema dimensional model consisting of FactSales and multiple dimension tables to support analytical reporting.
- Implemented Slowly Changing Dimension (SCD Type 2) for DimCustomer to preserve historical customer records.
- Developed SSIS Data Flow packages utilizing Lookup Transformations to validate business keys and maintain referential integrity during fact loading.
- Orchestrated ETL workflows through SSIS Control Flow to automate dimension loading prior to fact table processing.
- Ensured data consistency and integrity across customer, product, salesman, and sales fact tables within Microsoft SQL Server.

Tech Stack: SQL Server Integration Services (SSIS), Microsoft SQL Server, T-SQL, ETL, Data Warehousing, Star Schema, SCD Type 2

GitHub: <https://github.com/Aya-Abdel-Nasser/End-to-End-Sales-DataWarehouse-ETL>

SKILLS

Technical Skills

Programming Python, SQL

Data Engineering ETL/ELT, Data Pipelines, Data Modeling, Star Schema

Big Data & Streaming Apache Spark, PySpark, Apache Kafka

Cloud & Warehousing BigQuery, Microsoft Fabric, SQL Server, PostgreSQL

Orchestration: Apache Airflow, dbt

DevOps: Docker, Git, GitHub

Analytics: Power BI

Soft Skills

• Problem Solving • Analytical Thinking • Communication • Teamwork • Continuous Learning • Documentation